

Certified Predictability Horizons for Equivariant World Models

How far a world model can promise — tightly, and only with structure.

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Abstract

Scaling a world model lowers its average error but issues **no guarantee about any specific unseen situation, and none about how far into the future it can be trusted**. We give a different *kind* of result for **equivariant** latent world models: a **tight, computable certificate of the predictable horizon**. For a model whose encoder and latent predictor are G -equivariant with an orthogonal representation, the T -step rollout error is provably *constant over the entire orbit* of group-related situations (Theorem A), and — our central result — the *horizon* to which this holds is governed channel-by-channel by the predictor’s spectrum: a channel with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) = \Theta(\frac{1}{\lambda_j} \log \frac{1}{\epsilon})$, **and this is tight** — we prove a matching lower bound (Proposition 6) showing that any model that is only *approximately* equivariant is horizon-limited at exactly this rate, while a channel is certified to *unbounded* horizon **iff** it is conserved or invariant (the Noether hinge, Propositions 4–5). Two structural facts make the certificate exclusive to structure: the certificate is *equivalent* to equivariance (Lemma 2, the converse), so **no non-equivariant model possesses it at any scale**; and the existing single-shot certified-equivariance guarantees (robustness margins, equivariant conformal prediction) are exactly the $T=1$, ϵ -independent slice of this picture. Empirically, on a real physics-engine contact simulator (PushT) the certificate holds for a *learned* model to the floating-point floor while a $160\times$ parameter sweep of non-equivariant baselines never reaches it out of distribution; the certificate transfers exactly from the circle to non-abelian $SO(3)$ and — via frame averaging — to raw pixels at **no accuracy cost** relative to an unconstrained CNN. *Scale buys interpolation; structure buys a certified horizon.*

1. Introduction

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps**. Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance is known to give *some* per-situation guarantee. If a network commutes with a symmetry group G , its behaviour is constant across each group orbit — a fact used to certify adversarial robustness (orbit-constant classification margins) and to tighten conformal prediction sets (group-averaged nonconformity scores). But every such result is **single-shot**: it attaches a guarantee to one input, for one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor’s spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2)**. For an equivariant model the T -step rollout error is orbit-constant (Theorem A); relaxing to approximate equivariance, channel j with Lyapunov exponent λ_j is certified

to horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B, the *upper* bound). Our central new result is a **matching lower bound** (Proposition 6): an ϵ -approximately-equivariant model — even one with a *perfect* predictor — has orbit-error-variation exactly $\epsilon e^{\lambda T}$ on an expansive channel, so its certified horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**. Hence *approximate equivariance is horizon-limited; only exact equivariance or conservation reaches an unbounded horizon*.

2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ the model is equivariant. So **no non-equivariant model has the certificate at any size** — an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 places each conserved charge in its forced isotypic block, and Proposition 5 turns conservation into a long-horizon guarantee (T -step charge error $\leq T\eta$, linear, not the chaotic $e^{\lambda T}$). This unifies the configuration and horizon axes: the slow, certifiable subspace is the group-invariant subspace.
4. **Single-shot certified equivariance is our $T=1$ slice (§5).** We position the robustness/conformal guarantees as the one-step, resolution-free special case, and validate the multi-step picture on a real contact simulator, on non-abelian $\text{SO}(3)$, and on raw pixels — where frame averaging makes the certificate *accuracy-neutral*.

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

2. Setup and assumptions

A latent world model is an encoder $E: \mathcal{X} \rightarrow \mathcal{Z}$ and a predictor $f: \mathcal{Z} \times \mathcal{A} \rightarrow \mathcal{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \dots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T -step error is $\text{Err}_T(x) = \|\hat{z}_T - E(\Phi^T x)\|$. A group G acts on $\mathcal{X}$ (situations), $\mathcal{Z}$ (latents, via a representation ρ), and $\mathcal{A}$ (actions, via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g) E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g) f(z, a)$.
- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).
- **(A5) Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G -invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S \rangle$ and all x , $\text{Err}_T(w \cdot x) = \text{Err}_T(x)$. *Proof sketch.* By induction the rolled-out predictor $f^{(T)}$ is equivariant (composition of equivariant maps, Lemma 1 below), so $\hat{z}_T(w \cdot x) = \rho(w)\hat{z}_T(x)$ and the target $E(\Phi^T(w \cdot x)) = E(w \cdot \Phi^T x) = \rho(w)E(\Phi^T x)$ by (A3); subtract and use (A4). $\square$

Lemma 1 (composition closure). If (A1)–(A4) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$. Thus k **generator checks certify an exponentially large set**: k checks on S yield a guarantee over all of $\langle S \rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \rightarrow O(\mathcal{Z})$ act *freely* on an open $U \subseteq \mathcal{Z}$. If a predictor f 's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ , then f is equivariant on U . *Proof.* Fix z, g . For any probe c , freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant target on the orbit; orbit-constancy gives $\|f(z) - c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$ for *all* c , and two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. $\square$ With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance. **Hence no non-equivariant model possesses the certificate at any size** — the impossibility is a theorem, not an observation, and it is what the single-shot robustness/conformal literature (which proves only the forward direction) lacks.

3.2 The horizon axis, and its tightness (the central result)

Exactness is an idealization; real models are *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent map's Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} (λ_j a Lyapunov exponent).

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$|\text{Err}_T(w \cdot x) - \text{Err}_T(x)| \leq \sum_{\text{channels } j} c_j (m \epsilon_{\max} + T \delta) e^{\lambda_j T}, \quad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ ($\lambda_j > 0$), or $T_j = \infty$ ($\lambda_j \leq 0$). As $\epsilon_{\max}, \delta \rightarrow 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel with multiplier $a = e^\lambda$, $\lambda > 0$. There exist an exactly equivariant target Φ and a model that is ϵ -approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder that intertwines the dynamics along x 's trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$|\text{Err}_T(g \cdot x) - \text{Err}_T(x)| = \epsilon e^{\lambda T}.$$

Proof. $\text{Err}_T(x) = 0$ (model exact along x). At $g \cdot x$, linearity gives $f^T(E(g \cdot x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(E(x)) + a^T \epsilon u$ (using $f = \Phi$ equivariant and linear on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)\Phi^T(E(x))$ (by (A3) and encoder exactness off the single defect). Subtract and use (A4): $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. $\square$

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon's form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

Any model that is merely ϵ -approximately equivariant ($\epsilon > 0$) has a *finite* certified horizon on every expansive channel; only exact equivariance ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) reaches infinite horizon.

This is the horizon-domain companion of Lemma 2, and it sharpens *scale vs. structure*: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at an $\epsilon \approx 10^{-4}$ approximation level, never exact; §5), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so a single-step tie between augmentation and equivariance **must break over horizon** at the predicted T .

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \leq 0$ (unbounded-horizon) channels unidentified. The Noether hinge identifies them as the *conserved/invariant* ones, and ties the configuration axis to the horizon axis.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ : a scalar invariant (energy) in the trivial ($\ell=0$) block; a vector charge (angular momentum) in the $\ell=1$ block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory, not chosen.

Proposition 5 (conservation $\Rightarrow$ unbounded horizon). Let $Q : \mathcal{Z} \rightarrow W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \leq \eta$). Then the T -step *charge-value* prediction error satisfies $\|Q(\hat{z}_T) - Q(z_T)\| \leq T\eta$ — **linear in T** , never the chaotic $e^{\lambda T}$. So a conserved channel has $\lambda_Q \leq 0$ and, at $\eta=0$, is certified to all horizons. (The statement is about the charge value, and is exact under an equivariant symplectic discretization; the converse fails — a slow channel need not be conserved.)

Together: **the certified region is the coarse (ϵ large), invariant/conserved ($\lambda \leq 0$), low-composition ($|w|$ small) corner** — and the converse (Lemma 2) says this corner is reachable only with structure.

3.4 The certificate is a procedure

The certificate is *computable a priori*, without testing the unseen situation. Algorithm 1: (i) check (A1)–(A4) on the k generators to a residual $\epsilon_{\max}$; (ii) estimate the predictor Jacobian spectrum $\{\lambda_j\}$ at the query latent; (iii) report, for a target situation $w \in \langle S \rangle$, horizon T , resolution ϵ , whether (w, T, ϵ) lies in the certified region $\{|\Delta \text{Err}| \leq \epsilon\}$ implied by Theorem B / Proposition 6, escalating conserved channels to unbounded horizon via Proposition 5.

4. Experiments

All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports **INCONCLUSIVE** rather than loosen a threshold). We report the load-bearing subset; the certificate’s machinery is identical across them.

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$ compositions to $\sim 10^{-33}$ error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E2) The horizon staircase, and tightness. On a controlled latent with a tunable Lyapunov spectrum, the measured certified horizon recovers $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ to within 0.4% (slope 1.3–1.6 across seeds): chaotic channels are certified to 3–10 steps, slow/invariant channels to ≥ 90 . The recovery matches *both* Theorem B (upper) and Proposition 6 (lower), confirming the horizon is the true one, not merely attainable.

(E3) Approximate symmetry validates Proposition 6. Breaking the world’s symmetry by ϵ_{world} , the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$ –0.06 — exactly the crossing Proposition 6 predicts (where $\epsilon e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$ -equivariant, is horizon-limited as predicted.

(E4) Structure vs. scale. The equivariant model is orbit-flat (ratio 1.1–1.2); an $88\times$ -scaled non-equivariant model buys in-distribution interpolation (31–166 $\times$ better in-wedge, *beating* the equivariant model there) but its out-of-wedge error stays 10–155 $\times$ above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

(E5) The certificate on real contact dynamics. On PushT — a pymunk physics engine whose square arena makes $\text{SO}(2)$ scene rotation an exact dynamical symmetry — a *learned* $\text{SO}(2)$ -equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual $\sim 10^{-7}$) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a $160\times$ parameter sweep** ($1.7\text{k} \rightarrow 272\text{k}$) **reaches the equivariant floor out of distribution** (2.1–3.9 $\times$ above it, 3 seeds). The dynamics were not authored by us.

(E6) The certificate is not $\text{SO}(2)$ -specific, and transfers to pixels at no accuracy cost. It lifts to non-abelian $\text{SO}(3)$ on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a $7.4\times$ -smaller model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** — a plain CNN/MLP made exactly C_4 -equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN

on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing.

(E7) The certificate becomes task competence. Run through a G -equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a $4.3\times$ -larger non-equivariant controller degrades out of the training wedge.

5. Related work

Single-shot certified equivariance is our $T=1$ slice. The closest guarantee-bearing works live in robustness and conformal prediction. Equivariant classifiers have *orbit-constant margins* (the decision-boundary gradient norm is preserved across the orbit), giving uniform adversarial certificates (arXiv:2510.16171); invariance-aware randomized smoothing builds *orbit-based* certificates (arXiv:2211.14207); and Equivariantized Conformal Prediction (eCP, arXiv:2602.03986) group-averages the nonconformity score — frame averaging for distribution-free coverage. **All are single-shot** (one input, one prediction), **forward-only** (no converse), and have **no horizon, spectrum, or conservation axis**. They are exactly the $T=1$, ϵ -independent slice of our certificate: our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge to unbounded horizon (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al., 2021; arXiv:2112.03321) meta-learn conserved quantities inside the prediction loop, and Noether’s Razor (2024; arXiv:2410.08087) learns symmetries-as-conserved-quantities by Bayesian model selection. These *improve average prediction* by shrinking the hypothesis space; we instead *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which block must carry it. Guarantee versus average accuracy.

Jacobian-regularized world models. “Towards Unraveling and Improving Generalization in World Models” (arXiv:2501.00195) penalizes the latent-transition Jacobian to damp rollout error propagation — a heuristic for robustness. Theorem B is the provable version of the same intuition: read a per-channel certified horizon off the spectrum rather than regularize toward stability; Proposition 6 characterizes how that horizon shrinks under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRO-JEPA, UWM-JEPA) match a group structure to the latent and report strong zero-shot transfer; Theorem A explains *why* (the representation cancels inside the norm) and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic-Gaussian latent priors and their critics target a *distributional* (second-order) property; our certificate is a first-order, per-situation guarantee, and predicts precisely when a data-discovered latent anisotropy will fail to generalize (off the orbit).

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction). Our contribution is to (i) prove it *tight* for a *learned* latent world model, (ii) tie its unbounded-horizon subspace to the group-invariant subspace (Noether hinge), and (iii) make the forward direction’s converse a theorem (Lemma 2).

6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** The certificate is *exact* where G is a symmetry of the dynamics (orbital/conservative systems, free space, idealized manipulation) and *gracefully approximate* elsewhere, with a degradation that is now bounded on both sides (Theorem B and Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** Proposition 6 makes the horizon’s $\log(1/\epsilon)/\lambda$ *form* tight, but the constants c_j are not estimated and the isotypic refinement is asserted, not separately measured.

- **The Noether hinge’s forward direction is proved; its hypotheses are measured.** Proposition 5 assumes a Hamiltonian latent flow with a G -invariant Hamiltonian; the defect η is exact only under an equivariant symplectic discretization; the non-converse (slow $\nRightarrow$ conserved) is genuine.
 - **Scale and modality.** All experiments are 1–2-GPU. The exact flatness of the certificate transfers across modalities (structured state, $SO(3)$ point clouds, pixels), but absolute multi-step accuracy on raw pixels at this scale is poor for *every* architecture (an artifact of the anti-collapse-regularized JEPA latent, not of equivariance); a strong few-step pixel predictor at small scale is an architecture-agnostic open problem orthogonal to the certificate.
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7. Conclusion

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant subspace (Noether); and the whole region is reachable only with structure (the converse). The single-shot certified-equivariance guarantees the community already trusts are the one-step slice of this picture. *Scale buys interpolation; structure buys a certified horizon.*

References

- **Orbit-constant margins for equivariant networks** — orbit-invariant classification margin; single-shot, forward only. arXiv:2510.16171.
- **Equivariantized Conformal Prediction (eCP)** — group-averaged nonconformity score for orbit-wide coverage; single-shot, no horizon. arXiv:2602.03986.
- **Invariance-aware randomized smoothing** — orbit-based robustness certificates. arXiv:2211.14207.
- **Noether Networks** (Alet et al., 2021) — meta-learned conserved quantities for average prediction. arXiv:2112.03321.
- **Noether’s Razor** (2024) — learned symmetries-as-conserved-quantities via Bayesian model selection. arXiv:2410.08087.
- **Jacobian-regularized world models** — Jacobian penalty to damp rollout error propagation. arXiv:2501.00195.
- **Frame Averaging** (Puny et al., 2022) — Reynolds-operator construction making any backbone exactly equivariant; used here for the pixel certificate. arXiv:2110.03336.

Companion line. This paper is a focused extraction of the project’s *Certified World Models* program; the full experiment suite, the closed-loop clause, the discovery/generation re-framing, and the complete proofs appear in the companion manuscript `paper2_certified_world_models.md`.